

1 AL 600

¿CUÁNTOS N° DIVISIBLES

por 2, NI por 3, NI por 7?

$$\frac{600}{2} = 300 \quad \frac{600}{3} = 200$$

$$\frac{1}{7} 600 = 85,7$$

$$\frac{300}{3}$$

$$\begin{array}{r} 85 \\ \times 7 \\ \hline 595 \end{array} \quad \begin{array}{r} 80 \\ \times 0,5 \\ \hline 40 \end{array}$$

$$\frac{600}{6} = 100$$

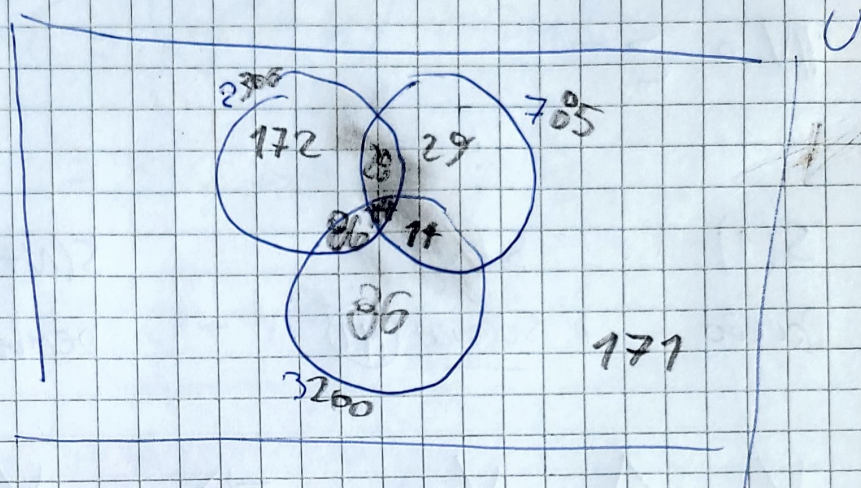
$$\frac{600}{14} = 42,85$$

$$\frac{600}{21} = 28,57$$

$$\frac{600}{42} = 14,28$$

$$300 + 86 + 14 + 29 = 429$$

$$600 - 429 = 171$$



PRINCIPIOS DE INDUCCIÓN MATEMÁTICA (PIM)

$$\forall n \in \mathbb{Z}^+$$

* HIPOTESIS

~~INDUCTIVA~~

INDUCTIVA

$S(1)$

PROUEBO

$S(n)$

SUPONGO $(H1)^*$

$S(n+1)$

DEMUESTRO

$$\underbrace{V \wedge V}_{\Rightarrow} V$$

$$\underbrace{V \Rightarrow V}_{\Rightarrow} V$$

CONCLUSIÓN : ES CIERTO QUE $3 | n^3 - n \quad \forall n \in \mathbb{Z}^+$

PROBAR " $n^3 - n$ ES DIVISIBLE POR 3 PARA TODO n POSITIVO

$$\underbrace{3 | n^3 - n}_{\text{CIRCULO}} \quad \forall n \in \mathbb{Z}^+$$

$$n^3 - n = 3 \cdot q$$

$\downarrow$
 $q \in \mathbb{Z}^+$

$$S(1) = 1^3 - 1 = 0 \quad 3 | 0 \quad V$$

$$\underline{S(n)} = \underbrace{n^3 - n}_{\text{CIRCULO}} = 3 \cdot q \quad (H1)$$

$$\underline{S(n+1)} = (n+1)^3 - (n+1) \stackrel{D}{=} 3 \cdot q_2$$

$$H^3 + 3H^2 + 3 \cdot H \cdot 1 - H - 1 = (H^3 - H) + 3H^2 + 3H$$

$$= 3 \cdot 0 + 3 \cdot H^2 + 3H =$$

$$3(0 + H^2 + H) =$$

$$3 \cdot 02 = \textcircled{V}$$

$$\Rightarrow S(1) \wedge S(N) \Rightarrow S(N+1)$$

$$V \quad 1 \quad \textcircled{V} \rightarrow V$$

$$LA \quad HI \quad ES \quad V$$

$$\textcircled{2} \quad 1+2+3 \dots + N \quad LA \quad SUMA \quad ES \quad S(N) = \frac{N(N+1)}{2}$$

$$S(4) = \frac{4(4+1)}{2} = \frac{20}{2} = 10 \quad \text{DEMOSTRADO POR REVERSO}$$

$$S(1) = \frac{1(1+1)}{2} = \frac{2}{2} = 1$$

$$S(N) = 1+2+3 \dots + N \quad \frac{N(N+1)}{2}$$

$$S(n+1) = \frac{(n+1) \cdot (n+1+1)}{2}$$

$$S(1) = \frac{1(1+1)}{2} = \frac{2}{2} = 1$$

$$S(n) = 1+2+3 \dots + n = \frac{n(n+1)}{2} \quad \text{HI}$$

$$S(n+1) = \underbrace{1+2+3 \dots + n}_{\downarrow} + n+1 = \frac{n+1(n+1+1)}{2}$$

$$\frac{n(n+1)}{2} + n+1 =$$

$$\frac{n(n+1) + 2(n+1)}{2} = \boxed{\frac{(n+1)(n+2)}{2}}$$

$$\textcircled{\checkmark} \Rightarrow \boxed{1+2+3+\dots+n = \frac{n(n+1)}{2}}$$

GAUSS

$$1+2+3 \dots 99+100$$

$$\underbrace{\hspace{10em}}_{101}$$

$$\begin{array}{r} 101 \\ \times 50 \\ \hline 5050 \end{array}$$

$$\frac{50 \cdot 100(100+1)}{2}$$

$$\textcircled{3} \quad S(1) = 1 = 1^2$$

$$S(7) = 1+3+5+7 = 16 =$$

$$\textcircled{3} \quad \text{IMPAR} \quad 2 \cdot N - 1$$

$$1+3+5+\dots+(2N-1) = N^2$$

$$S(4) = 1+3+5+7 = 4^2 = 16$$

$$S(4) = 1+3+5+\dots+(2 \cdot 4 - 1) = 4^2$$

✓

$$S(n+1) = 1+3+5+\dots+(2n-1) + 2(n+1)-1 = (n+1)^2$$

$$n^2 + 2(n+1) - 1 = (n+1)^2$$

$$n^2 + 2n + 2 - 1 = (n+1)^2$$

$$\underbrace{n^2 + 2n + 1}_{=} = (n+1)(n+1) = \underbrace{n^2 + n + n + 1}_{=} = n^2 + 2n + 1$$

✓

$$\textcircled{4} \quad 1 \cdot 2 + 2 \cdot 3 + 3 \cdot 4 + \dots + N(N+1) = \frac{N \cdot (N+1)(N+2)}{3}$$

$$S(N) =$$

$$S(4) = 1 \cdot 2 + 2 \cdot 3 + 3 \cdot 4 + \underset{\substack{\downarrow \\ (4 \cdot (4+1))}}{4 \cdot 5} = \frac{4 \cdot (4+1)(4+2)}{3}$$

40

$$\frac{120}{3}$$

$$40$$

$$S(1) = 1 \cdot 2 = \frac{1 \cdot (1+1)(1+2)}{3}$$

$$\frac{6}{3} = 2$$

$$S(N) = 1 \cdot 2 + 2 \cdot 3 + 3 \cdot 4 + \dots + N(N+1) = \frac{N(N+1)(N+2)}{3}$$

$$S(N+1) = \underbrace{1 \cdot 2 + 2 \cdot 3 + 3 \cdot 4 + \dots + N(N+1)}_{S(N)} + (N+1)(N+1+1) = \frac{(N+1)(N+1+1)(N+1+2)}{3}$$

$$\frac{N(N+1)(N+2)}{3} + (N+1)(N+1+1)$$

$$\frac{n(n+1) \cdot (n+2)}{3} + (n+1) \cdot (n+2)$$